Dividing polynomials

1

a $\frac{3}{4}$ **b** $\frac{7v}{71}$ **c** $\frac{7x}{3}$ **d** $2p^4$

e $\frac{10n}{3m}$

2

a False **b** x^6

3

a $\frac{z}{10}+1$ **b** $5x^2+4x^4$

c $6x^3-4x^2+8$ **d** $4x^3-7x^2+3x+6$

4

a (x^3-2) units **b** $(3x^3+2x^2+5x+7)$ units

5 $(26n^2+22n+58)$ units

6

a $\frac{3x-8}{6}$ **b** -1 **c** -3 **d** $m+10$

e $m-1$ **f** $-(a+9)$ **g** $\frac{x+2}{10}$ **h** $\frac{x+4}{x-5}$

i $x+5$ **j** $\frac{2(x+11)}{5}$ **k** $\frac{2}{x}$ **l** $-\frac{5}{2x}$

m $\frac{m-1}{2}$ **n** $x-5$ **o** $\frac{2x+3}{4x-5}$

7 $y-1$

Additional questions

8

a $30ab^2c$ **b** $2x$

c $14mn^3$, mn, and $\frac{3}{2}m^2$

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9

a x^6

b 1

10 Lawrence subtracted both the coefficients and the exponents of the variable. However, when dividing by a monomial the coefficients should be divided and the exponents should be subtracted. Marika, on the other hand, divided both the coefficients and the exponents when she should have divided the coefficients and subtracted the exponents.

11

a False. We can't create a coefficient of zero when dividing a single term of a polynomial by a monomial so no terms will eliminate.

b False. If you divide by a constant monomial the degree will not change.

c True. Suppose the monomial has degree n and the term has degree n. Then $ax^n/bx^n = a/b$ will be a constant.

d False. If the degree of the monomial is larger than the degree of any of the terms in the polynomial the result will not be a polynomial.

12 In the first expression we are dividing all terms in the numerator by 5. In the second expression only the constant term at the end is being divided by 5.

13

a Each term of the resulting polynomial will have a fractional coefficient.
$\frac{x^2+2}{4} = \frac{1}{4}x^2 + \frac{1}{2}$

b The terms will have negative exponents and the expression will no longer be a polynomial.
$\frac{x^2+2}{x^3} = x^{-1} + 2x^{-2} = \frac{1}{x} + \frac{2}{x^3}$

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